The sixth Schur number is at least 538

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November 2026. doi:10.5281/zenodo.23937655

Abstract

We give an explicit partition of $\{1, \ldots, 538\}$ into six sum-free sets. This shows $S(6) \geq 538$ and improves the bound $S(6) \geq 537$ of Chen, Zhu and Long. It also gives $R_6(3) \geq 540$ for the Ramsey number of the triangle in six colours, and known recursive constructions give $S(8) \geq 5382$, $S(11) \geq 204588$ and $S(13) \geq 2045882$. The partition of $\{1, \ldots, 538\}$ is checked directly and in Lean 4.

1. The result

A set A of positive integers is *sum-free* when there are no $x, y \in A$ with $x + y \in A$; the elements x and y may be equal. The Schur number $S(k)$ is the largest n such that $\{1, \ldots, n\}$ is the union of k sum-free sets. Schur proved that $S(k)$ is finite [Sch16].

Fredricksen and Sweet gave $S(6) \geq 536$ [FS00], and Chen, Zhu and Long gave $S(6) \geq 537$ [CZL26]. The latter is the largest lower bound in the sources that we read on 11 November 2026.

Theorem 1. $S(6) \geq 538$.

Proof. Let W be the word of 538 letters in Figure 1, and for $i = 1, \ldots, 6$ let C_i be the set of positions n such that letter n of W is i. These six sets partition $\{1, \ldots, 538\}$, and their sizes are $131, 85, 58, 67, 63, 134$. There are $269^2 = 72361$ pairs (x, y) with $1 \leq x \leq y$ and $x + y \leq 538$. For each pair, the letters at positions x, y and $x + y$ are not all the same. So each C_i is sum-free. □

Figure 1. The word W. Each row gives the position of its first letter.

```
  1  1214532151 4154552214 1213122344 2151413435 2315124412
 51  2541313532 2515141312 2334213142 5552231312 3452214421
101  5133531314 1552443221 3122432215 1514353321 4121614245
151  3215141545 5121412163 6214121614 5641516161 6266661413
201  1656465161 4166626616 1616266661 6661666666 6666666636
251  6161666666 6666666666 6666666666 6666666663 6666666166
301  6666161666 6662616666 6626661666 6564656131 4166662616
351  1615646541 6121412636 1214121564 5141512354 2416121412
401  3351415151 2234221312 2344255141 3135231512 4412254321
451  3132255524 1312433221 3141515223 5313145221 4421513253
501  4314151244 3221312141 2255451415 12354121
```

Let $R_k(3)$ be the least N such that every colouring of the edges of the complete graph K_N with k colours has a monochromatic triangle.

Corollary 2. $R_6(3) \geq 540$.

Proof. We use the classical construction behind $S(k) \leq R_k(3) - 2$ [FS00, (2)]. Take the vertices $0, 1, \ldots, 538$ of K_{539}, and give the edge $\{a, b\}$ the colour i with $|a - b| \in C_i$. A monochromatic triangle $a < b < d$ would put $x = b - a$, $y = d - b$ and $x + y = d - a$ in one set C_i. This is not possible, because C_i is sum-free. So this colouring with six colours has no monochromatic triangle, and neither has its restriction to fewer vertices. So $R_6(3) > 539$. □

Corollary 2 improves the bound $R_6(3) \geq 539$ of [CZL26]. The survey [Rad26, Table XIa] lists the earlier bound $R_6(3) \geq 538$ of [FS00].

Corollary 3. $S(8) \geq 5382$, $S(11) \geq 204588$ and $S(13) \geq 2045882$.

Proof. We use the recurrences $S(k+2) \geq 10\,S(k) + 2$ [BBG26, Section IV] and $S(k+5) \geq 380\,S(k) + 148$ [ACP22, inequality (6)]. With Theorem 1 they give

$$S(8) \geq 10 \cdot 538 + 2 = 5382, \qquad S(11) \geq 380 \cdot 538 + 148 = 204588,$$

$$S(13) \geq 10 \cdot 204588 + 2 = 2045882. \qquad \square$$

These improve the bounds 5372, 204208 and 2042082 of [CZL26], which come from the same recurrences applied to $S(6) \geq 537$.

2. Checks

The following Python program holds the only computation that Theorem 1 needs. With `W` the word of Figure 1, as one string, it prints `538 ['1', '2', '3', '4', '5', '6'] []`. The last item is the list of pairs (x, y) with letters at x, y and $x + y$ all the same.

```
w = " " + W
bad = [(x, y) for x in range(1, 270) for y in range(x, 539 - x)
       if w[x] == w[y] == w[x + y]]
print(len(W), sorted(set(W)), bad)
```

The existence of the partition is also checked in Lean 4, version v4.35.0-rc3. The file `lean/SchurSix538.lean` imports Lean core only and states, in the namespace `SchurWitness538`:

```
public theorem schur_six_538 :
    ∃ c : Nat → Fin 6, ∀ x y : Nat, 1 ≤ x → 1 ≤ y → x + y ≤ 538 →
      ¬ (c x = c y ∧ c y = c (x + y))
```

For the witness in the file, $c(n) + 1$ is letter n of W for $1 \leq n \leq 538$ (theorem `colouring538_val`, which is private to the file). So the sets $\{n \in \{1, \ldots, 538\} : c(n) = i\}$, for i in `Fin 6`, are the sets C_{i+1} of Theorem 1. Lean does not define $S(k)$ or $R_k(3)$: the step from the partition to $S(6) \geq 538$ is the definition of $S(6)$, and Corollaries 2 and 3 are not formalized. The theorem uses the axioms `propext` and `Quot.sound`. This file has no `sorry` and does not use `native_decide`. The file `lean/Challenge.lean` states the same theorem with `sorry` on purpose, for the comparator of the Lean package; the proof does not use it.

Corollary 3 rests on two preprints, but its three partitions are also checked directly, and this check uses only the tables of those preprints. The program `check_recursive_bounds.py` in the repository applies [BBG26, Table II] to W (8 sets, $\{1, \ldots, 5382\}$), [ACP22, Table 7] to W (11 sets, $\{1, \ldots, 204588\}$), and [BBG26, Table II] to the partition with 11 sets (13 sets, $\{1, \ldots, 2045882\}$). It finds that each set is sum-free.

3. Data availability

The word, the Lean package and the program `check_recursive_bounds.py` are at https://github.com/mysticflounder/schur-538, under the license CC BY 4.0. The file `schur6-538.word` has the letters of W and one final newline, which must be removed to get the output stated in Section 2. Its SHA-256 is

`9e149c007b778655afd199685ebcc76260604a44f4c322d8aa9a1d96afb71d20`,

the SHA-256 of `lean/SchurSix538.lean` is

`ef4fa1757ff30ed191f7eb1af24af288ab0a67726ee52b108355402789a63b81`,

and the SHA-256 of `check_recursive_bounds.py` is

`aa7620b4dceb9788fa2a1babca998b01f2bcf266ac1fc9aaeda896397f25ff0c`.

The repository also holds a second such word, `schur6-538-b.word`, which differs from W in 37 letters. The Lean theorem does not use it.

Use of AI tools

The word was found by a computer program whose development the author directed. AI models (Anthropic Claude and OpenAI GPT models) wrote the program. The computer checks, the Lean formalization and this note were produced with the AI system Claude (Anthropic; model Claude Opus 5.5) under the direction of the author. Separate agents of the same model audited the word, the Lean theorem and the partitions of Corollary 3 with their own code. The author has checked the content and takes responsibility for it.

References

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